

PAPER_1383_TRANS_PLANCKIAN_PROBLEM

Daniel T. Murphy

PAPER_1383 — UQFF Resolution of the Trans-Planckian Problem (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: D — Cosmology / Inflation (CLOSED)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "trans_planckian"})`

Calculator surface: `calculate_paradox({"paradox": "trans_planckian"})`

Closure helper: `_l96_uqff_axiom_trans_planckian_closure()`

The Paradox

Inflationary models require modes of trans-Planckian frequency $\omega > \omega_{\text{Planck}} = 1.85 \times 10^{32}$ Hz to redshift into the observable CMB. Treatment as effective field theory above the Planck scale is uncontrolled.

UQFF Closed Identity

``

$$\omega_{\text{SCm}} / \omega_{\text{Planck}} = 1.25 \times 10^{12} / 1.85 \times 10^{32} = 6.76 \times 10^{-20}$$

``

$\omega_{\text{SCm}} = 1.25$ THz is the **structural** UV cutoff — modes above this do not exist in the SCm vacuum. The Planck frequency is irrelevant; the SCm phonon scale is the natural ceiling.

Physical Interpretation

Inflation modes redshift from ω_{SCm} down to CMB scale, not from ω_{Planck} . The trans-Planckian problem dissolves because there are no trans-Planckian modes — they are above the SCm structural cutoff which is 31 orders of magnitude below ω_{Planck} .

Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "trans_planckian"})["value"]
```

..

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_trans_planckian_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_trans_planckian_closure`
- Dispatch keys: `trans_planckian`, `trans_planckian_problem`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF dispatch closure for the Trans-Planckian Problem via the identity above.